

Network Address Translation (Part 2)

- Topics covered:
 - More About Static NAT
 - Dynamic NAT
 - Dynamic PAT

More About Static NAT

Review: Static NAT

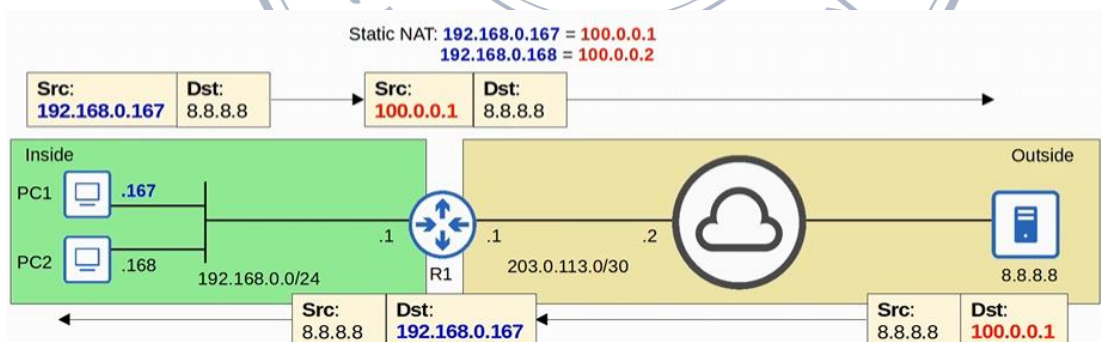
Static NAT uses permanent **one-to-one mappings** between private and public IP addresses.

Configured mappings:

1. 192.168.0.167 → 100.0.0.1
2. 192.168.0.168 → 100.0.0.2

When traffic leaves the internal network, the router translates the **source IP address**.

Example:



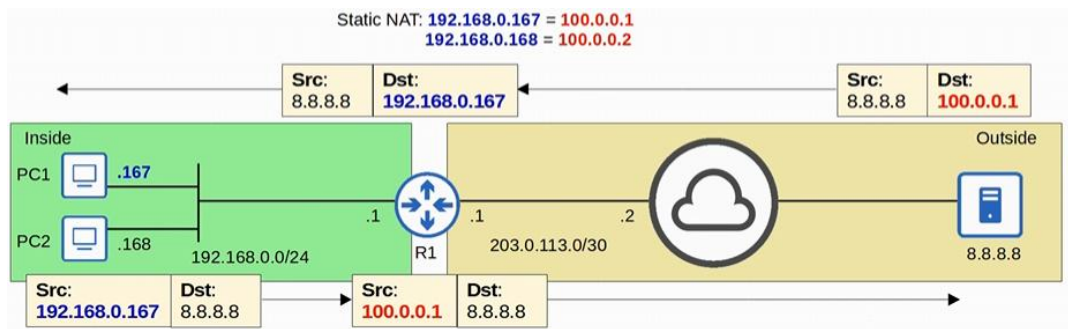
Static NAT Works in Both Directions

The one-to-one mapping is not limited to inside-to-outside communication.

External hosts can also initiate communication using the inside global address.

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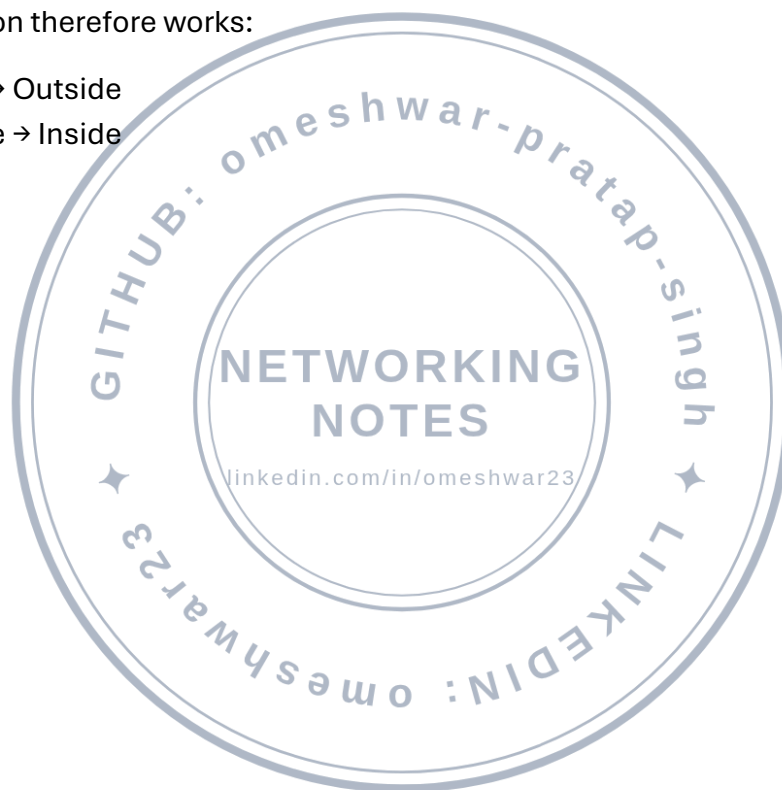
Outside Host Initiates Communication



The router translates the destination address from **100.0.0.1** to **192.168.0.167** and forwards the packet to PC1.

Communication therefore works:

1. Inside → Outside
2. Outside → Inside



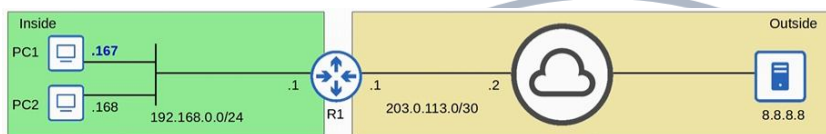
Dynamic NAT

How Dynamic NAT?

Unlike Static NAT, Dynamic NAT does **not** require manually mapping each private IP address to a specific public IP address.

Instead, the router:

1. Automatically creates mappings when needed.
2. Removes mappings when they are no longer needed.



Dynamic NAT Working

Cisco IOS uses two components:

1. ACL
2. NAT Pool

ACL Identifies Traffic

The ACL is **not** used to block traffic.

Instead, it simply identifies which traffic should be translated.

1. If a packet's source IP is permitted by the ACL: **NAT translates the source IP.**
2. If the source IP is denied: **NAT does not translate the source IP.**

A denied packet is **not dropped** because the ACL is **not** applied to an interface using ip access-group.

NAT Pool

A NAT pool defines the available **Inside Global** addresses.

Example: ACL 1 permits **192.168.0.0/24**

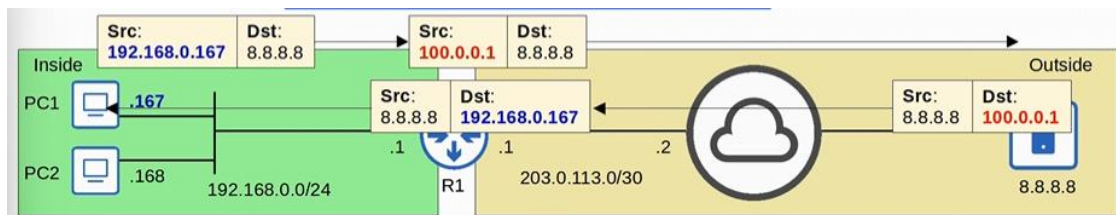
POOL1 contains: 100.0.0.1 to 100.0.0.10

```
On R1:
ACL 1: permit 192.168.0.0/24
      deny any
POOL1: 100.0.0.1 to 100.0.0.10
If a packet with a source IP permitted by ACL 1 arrives,
translate the source IP to an address from POOL1.
```

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Dynamic Translation Process

PC1 to communicate to 8.8.8.8



Unlike Static NAT, the router creates this mapping automatically when the packet arrives.

Important ACL Behavior

If a source IP is **not permitted** by the ACL: The packet is **not translated**.

This **does not** mean the packet is dropped, The ACL is only identifying which traffic should receive NAT.

One-to-One Mapping Still Exists

Although Dynamic NAT creates mappings automatically, they are still:

One Inside Local IP & One Inside Global IP

Each public address can be used by only one internal host at a time.

NAT Pool Exhaustion

If every address in the NAT pool is already assigned, no additional translations can be created.

This condition is called: **NAT Pool Exhaustion**

Example

Source IP	Translated Source IP
192.168.0.167	100.0.0.1
192.168.0.168	100.0.0.2
192.168.0.100	100.0.0.3
192.168.0.12	100.0.0.4
192.168.0.28	100.0.0.5
192.168.0.56	100.0.0.6
192.168.0.202	100.0.0.7
192.168.0.221	100.0.0.8
192.168.0.116	100.0.0.9
192.168.0.188	100.0.0.10
192.168.0.98	No address available! Router will drop the packet

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No address available, Router drops the packet.

The host cannot access outside networks until a public address becomes available.

Address Reuse

Suppose **192.168.0.167** stops communicating, Its translation eventually times out.

Then: 192.168.0.98 → 100.0.0.1

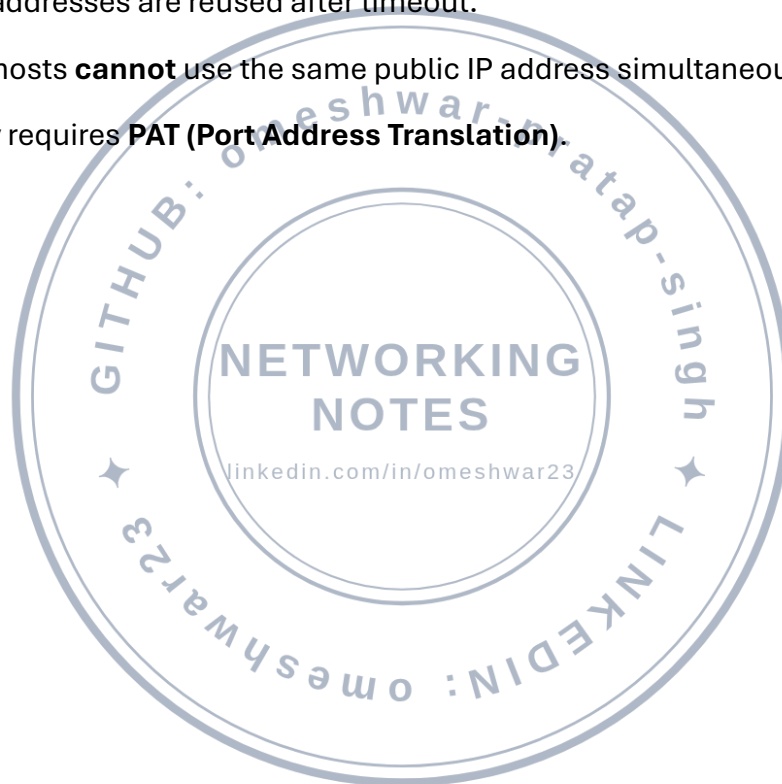
The same public IP address is reused by another host.

Unlike Static NAT:

- Dynamic mappings are temporary.
- Public addresses are reused after timeout.

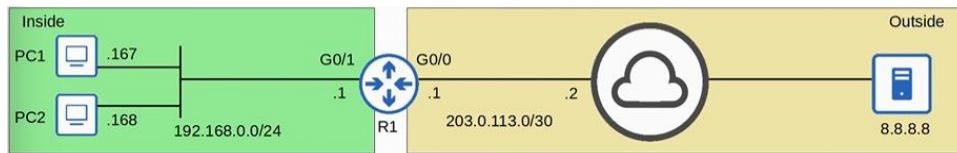
However, two hosts **cannot** use the same public IP address simultaneously.

That capability requires **PAT (Port Address Translation)**.



Dynamic NAT Configuration

How To Configure Dynamic NAT



1. Configure Inside and Outside Interfaces

Same as Static NAT.

```
R1(config)#int g0/1
R1(config-if)#ip nat inside
```

Define the 'inside' interface(s) connected to the internal network.

```
R1(config-if)#int g0/0
R1(config-if)#ip nat outside
R1(config-if)#exit
```

Define the 'outside' interface(s) connected to the external network.

2. Configure ACL

```
R1(config)#access-list 1 permit 192.168.0.0 0.0.0.255
```

Define the traffic that should be translated.
*Traffic permitted by this ACL will be translated.

Traffic permitted by this ACL will be translated.

```
access-list 1 permit 192.168.0.0.0.0.0.255
```

This identifies all traffic from **192.168.0.0/24**.

3. Configure NAT Pool

```
R1(config)#ip nat pool POOL1 100.0.0.0 100.0.0.255 prefix-length 24
```

Define the pool of inside global IP addresses.
*Instead of **prefix-length 24**, you can use **netmask 255.255.255.0**

Instead of prefix-length, netmask may also be used.

IOS verifies that the first and last addresses belong to the same subnet.

If they are not, the command is rejected.

4. Map ACL to NAT Pool

```
R1(config)#ip nat inside source list 1 pool POOL1
```

Configure dynamic NAT by mapping the ACL to the pool.

This tells R1:

1. Match traffic using ACL 1.
2. Translate using addresses from POOL1.

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Verifying Dynamic NAT

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 100.0.0.1:3       192.168.0.167:3   8.8.8.8:3          8.8.8.8:3
udp 100.0.0.1:58685    192.168.0.167:58685 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.1          192.168.0.167     ---               ---
icmp 100.0.0.2:3       192.168.0.168:3   8.8.8.8:3          8.8.8.8:3
udp 100.0.0.2:49536    192.168.0.168:49536 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.2          192.168.0.168     ---               ---
```

Explanation

Each inside-to-outside mapping has:

- One mapping entry.
- Additional entries whenever the mapping is actively used.

For each host: Mapping Entry + ICMP Translation + UDP Translation

Translation Timeouts

The UDP and ICMP translation entries are automatically removed after about one minute, The original Dynamic NAT mapping remains much longer.

Default timeout: **24 hours**

Every time the mapping is used, its timer resets.

The timeout values can be changed, but that configuration is not required for the CCNA.

```
! below is about 1 minute later

R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
--- 100.0.0.1          192.168.0.167     ---               ---
--- 100.0.0.2          192.168.0.168     ---               ---
```

Clearing Dynamic NAT Entries

Although the translation table resembles the Static NAT table, these mappings are **dynamic**.

Therefore: clear ip nat translation *

removes the Dynamic NAT entries because they are not permanent.

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Verifying Dynamic NAT

```
R1#show ip nat statistics
Total active translations: 6 (0 static, 6 dynamic; 4 extended)
Peak translations: 6, occurred 00:00:30 ago
Outside interfaces:
  GigabitEthernet0/0
Inside interfaces:
  GigabitEthernet0/1
Hits: 32 Misses: 0
CEF Translated packets: 20, CEF Punted packets: 12
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list 1 pool POOL1 refcount 6
  pool POOL1: netmask 255.255.255.0
    start 100.0.0.0 end 100.0.0.255
    type generic, total addresses 256, allocated 2 (0%), misses 0
[output omitted]
```

Explanation

Total active translations: 6

- Static translations: **0**
- Dynamic translations: **6**
- Extended translations: **4**

The extended entries are the temporary UDP and ICMP translation entries that are automatically removed after about one minute.

For the CCNA, the details of extended entries are not required.

The important information is that this command confirms the dynamic mapping:

ACL 1 → POOL1

For Dynamic NAT, the two verification commands to know are:

1. show ip nat translations
2. show ip nat statistics

Dynamic PAT

Port Address Translation

PAT is also known as **NAT Overload**.

PAT translates:

1. IP address
2. Port number (when necessary)

The purpose of translating port numbers is to allow **many internal hosts to share a single public IP address**.

TCP and UDP port numbers are **16 bits**, providing over **65,000** possible port numbers.

The router keeps track of communication by recording which:

- Inside Local address
- Inside Global address
- Port number

belong to each communication flow.

PAT Operation



Both PCs send DNS requests to **8.8.8.8**.

Initial packets:

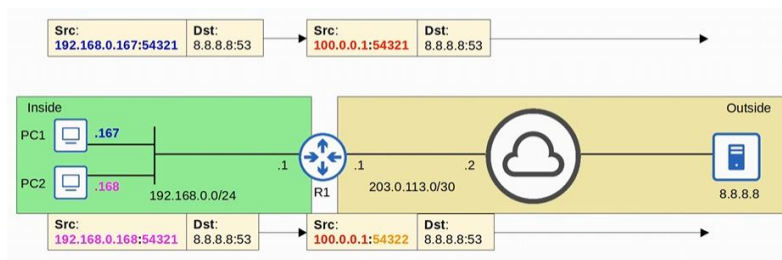
PC1	<table><tr><td>Src:</td><td>Dst:</td></tr><tr><td>192.168.0.167:54321</td><td>8.8.8.8:53</td></tr></table>	Src:	Dst:	192.168.0.167:54321	8.8.8.8:53
Src:	Dst:				
192.168.0.167:54321	8.8.8.8:53				

PC2	<table><tr><td>Src:</td><td>Dst:</td></tr><tr><td>192.168.0.168:54321</td><td>8.8.8.8:53</td></tr></table>	Src:	Dst:	192.168.0.168:54321	8.8.8.8:53
Src:	Dst:				
192.168.0.168:54321	8.8.8.8:53				

Both hosts happened to choose the same random source port.

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Translation by R1

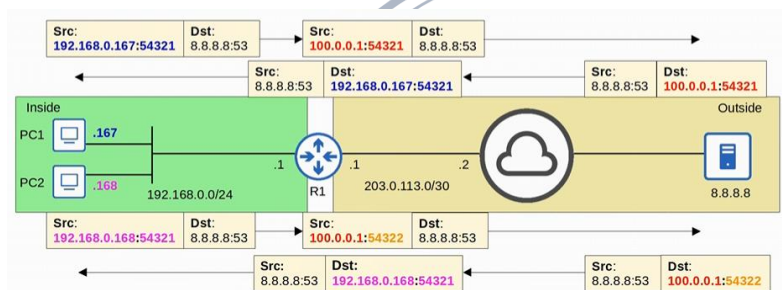


PC1 keeps port **54321**.

PC2's source port becomes **54322**.

The router changes the second port so that each communication flow remains unique.

Reply Traffic



Because each communication flow has a unique port number, R1 knows which reply belongs to which PC.

When Port Translation Is Not Needed

If different hosts already choose different source ports, R1 does **not** translate the port numbers, It simply uses the existing source ports to track each communication flow.

Why PAT Is Important

PAT allows:

1. Many inside hosts
2. To share one public IP address

Because of this, PAT greatly conserves public IPv4 addresses and is widely used.

Among:

1. Static NAT
2. Dynamic NAT
3. PAT

PAT is the most commonly used.

Dynamic PAT Configuration

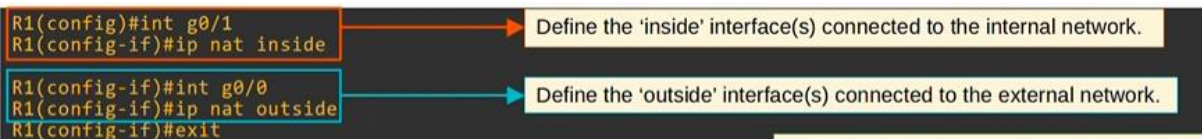
Configuring PAT (Using a NAT Pool)

Configuration is almost identical to Dynamic NAT.

The only difference is the **overload** keyword.

1. Configure Inside and Outside Interfaces

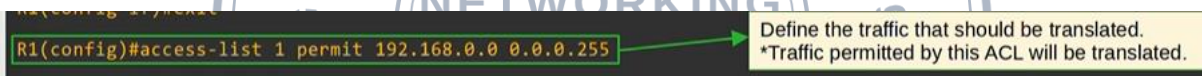
Same as previous NAT configurations.



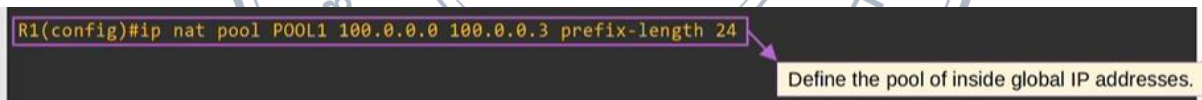
2. Configure ACL

Traffic permitted by the ACL will be translated.

```
access-list 1 permit 192.168.0.0 0.0.0.255
```



3. Configure NAT Pool



Although the address range is smaller, the prefix length is still **24**.

The command is valid because all addresses belong to the same subnet.

4. Enable PAT



```
ip nat inside source list 1 pool POOL1 overload
```

The overload keyword enables PAT.

After That:

PC1 and PC2 both send dns request



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Verifying PAT

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
udp 100.0.0.1:63925    192.168.0.167:63925 8.8.8.8:53         8.8.8.8:53
udp 100.0.0.1:59549    192.168.0.168:59549 8.8.8.8:53         8.8.8.8:53
R1#show ip nat statistics
Total active translations: 2 (0 static, 2 dynamic; 2 extended)
Peak translations: 2, occurred 00:00:03 ago
Outside interfaces:
  GigabitEthernet0/0
Inside interfaces:
  GigabitEthernet0/1
Hits: 4 Misses: 0
CEF Translated packets: 0, CEF Punted packets: 4
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 3] access-list 1 pool POOL1 refcount 2
  pool POOL1: netmask 255.255.255.0
    start 100.0.0.0 end 100.0.0.3
    type generic, total addresses 4, allocated 1 (25%), misses 0
```

Explanation

Unlike Dynamic NAT: No permanent one-to-one mapping entries appear.

This is because PAT creates:

- Many Inside Local addresses
- One Inside Global address

Both PCs use **100.0.0.1**.

Since they already selected different source ports, R1 did not need to translate the ports.

The router tracks:

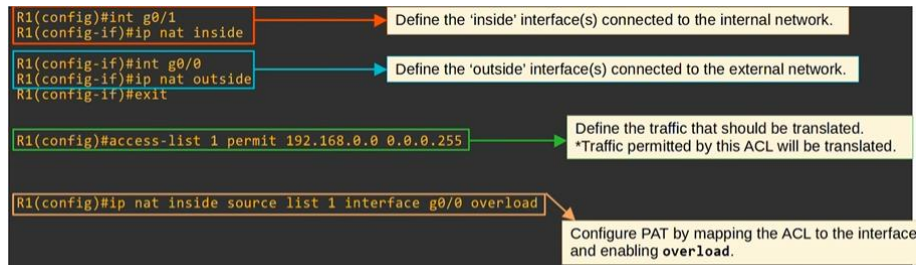
1. PC1 using port **63925**
2. PC2 using port **59549**

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Configuring PAT Using the Router's Interface

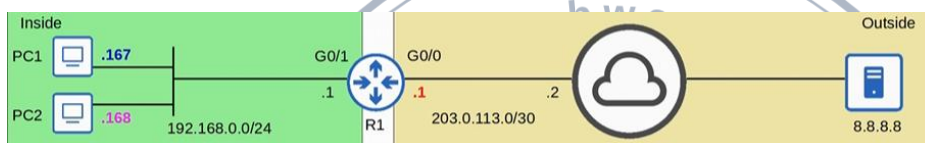
Instead of using a NAT pool, PAT can use the router's outside interface address.

Configuration:

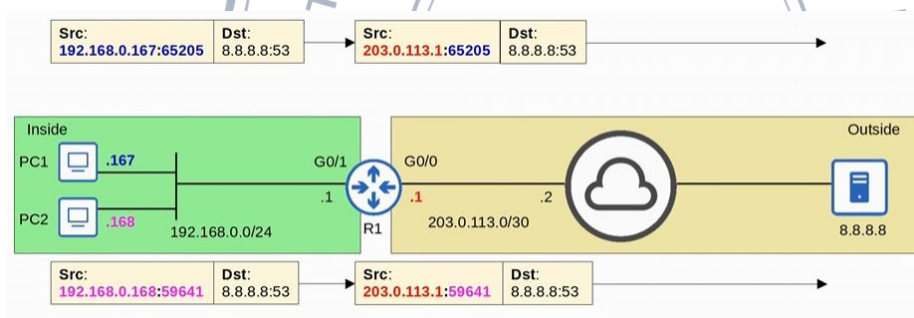


The router translates packets to the IP address of **G0/0**.

PAT Using the Interface Address



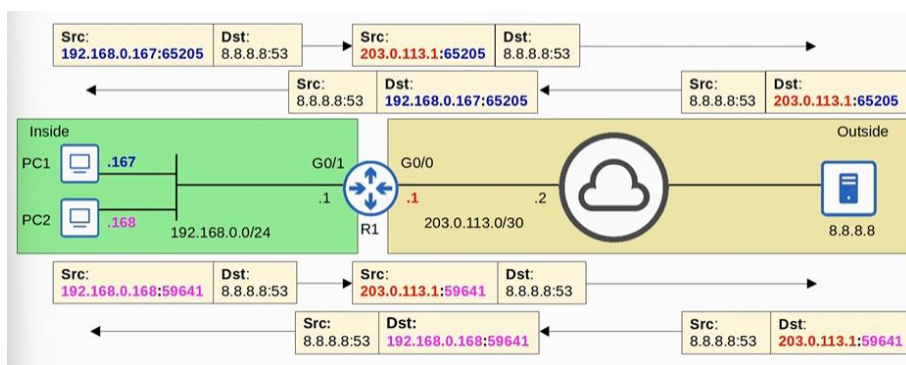
PC1 and PC2 both send dns request



Both hosts use the same public IP: **203.0.113.1**

Because their source ports are already different, R1 keeps the original port numbers.

Reply Traffic



The unique port numbers allow R1 to return each reply to the correct host.

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Verification

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
udp  203.0.113.1:65205  192.168.0.167:65205 8.8.8.8:53         8.8.8.8:53
udp  203.0.113.1:59641  192.168.0.168:59641 8.8.8.8:53         8.8.8.8:53
R1#show ip nat statistics
Total active translations: 2 (0 static, 2 dynamic; 2 extended)
Peak translations: 2, occurred 00:36:30 ago
Outside interfaces:
  GigabitEthernet0/0
Inside interfaces:
  GigabitEthernet0/1
Hits: 12 Misses: 0
CEF Translated packets: 0, CEF Punted packets: 12
Expired translations: 4
Dynamic mappings:
-- Inside Source
[Id: 4] access-list 1 interface GigabitEthernet0/0 refcount 2
```

Commands Summary

```
R1(config)# ip nat pool pool-name start-ip end-ip prefix-length prefix-length
R1(config)# ip nat pool pool-name start-ip end-ip netmask subnet-mask
R1(config)# ip nat inside source list access-list pool pool-name
R1(config)# ip nat inside source list access-list pool pool-name overload
R1(config)# ip nat inside source list access-list interface interface overload
```

